

Computer Science Program Summery

The completion of the Computer Science program's course requirements and the creation of an ePortfolio are crucial to showcasing my abilities, establishing my career objectives, and making the move to a lucrative software engineering career. This capstone project demonstrates my development from a student programmer to a professional software engineer with a broad range of problem-solving skills. I have two significant accomplishments that show my development. After finishing CS 250 (Software Development Lifecycle), I was exposed to the Agile process and learned the value of communicating clearly with stakeholders that have different technical backgrounds.

My improved Pirate Intelligent Agent artifact demonstrates my expertise in cutting-edge algorithmic techniques including priority queue implementations and reinforcement learning improvements within data structures and algorithms. My ability to apply theoretical ideas to real-world issues is demonstrated by the 28.8% improvement via priority-based experience replay.

The Android inventory management app and a MongoDB-powered animal shelter dashboard demonstrate my proficiency with databases and software engineering. These two projects demonstrate my versatility as a full-stack developer, encompassing everything from complicated database aggregations to mobile application design. This demonstrates how I progressed from academic research to enterprise-scale solutions, which aligns nicely with my professional experience in cloud infrastructure and automated testing frameworks.

From implementing BCrypt password hashing in the Android app to secure database connections and data verification in the dashboard app, security has consistently been a focus of my innovations.

This portfolio's three artifacts work well together to highlight my whole skill set as a software developer. These artifacts each reflect a distinct facet of software development: full-stack web development with database considerations, mobile app development with security considerations, and the use of sophisticated algorithms with performance enhancements. Together with my professional background in creating automated testing frameworks and Kafka streaming platforms, these artifacts show that I am prepared to contribute right away to any software development team while still learning and developing in this ever-evolving sector.